

Solutions & Notes of Homework 6

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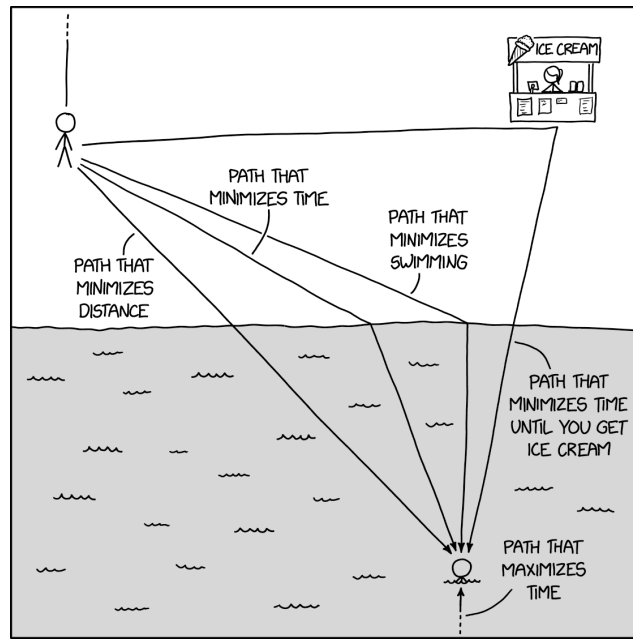


Figure 1: Picture from <https://matthbeck.github.io/quotes.html>, originally from <https://xkcd.com/>.

Problem 1

In $\mathbb{R}^4$, let

$$U = \text{span}((1, 1, 0, 0), (1, 1, 1, 2)).$$

Find $u \in U$ such that $\|u - (1, 2, 3, 4)\|$ is as small as possible.

✓ **Solution:** $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \\ 0 & 2 \end{pmatrix}$, $\dim C(A) = 2$. Thus

$$u = A(A^T A)^{-1} A^T (1, 2, 3, 4)^T = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} \frac{7}{10} & -\frac{1}{5} \\ -\frac{1}{5} & \frac{1}{5} \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 \end{pmatrix} (1, 2, 3, 4)^T = \left(\frac{3}{2}, \frac{3}{2}, \frac{11}{5}, \frac{22}{5} \right)^T.$$

! **Note 1.1:** You can also use Gram-Schmidt procedure to calculate the orthonormal basis

$$e_1 = \frac{1}{\sqrt{2}}(1, 1, 0, 0), \quad e_2 = \frac{1}{\sqrt{5}}(0, 0, 1, 2),$$

and use equation 6.57 (i) to find the answer.

Problem 2

Find $p \in \mathcal{P}_3(\mathbb{R})$ such that $p(0) = 0, p'(0) = 0$, and

$$\int_0^1 |2 + 3x - p(x)|^2 dx$$

is as small as possible.

✓ **Solution:** $p(0) = 0, p'(0) = 0$ implies $p(x) = ax^3 + bx^2 (a, b \neq 0)$. We can easily verify that all $p(x)$'s form a subspace of $\mathcal{P}_3(\mathbb{R})$, and x^2, x^3 forms a basis of it. Apply Gram-Schmidt procedure to x^2, x^3 and we get an orthonormal basis e_1, e_2 :

$$e_1 = \frac{x^2}{\sqrt{\int_0^1 (x^2)(x^2) dx}} = \sqrt{5}x^2,$$

$$f_2 = x^3 - \frac{\int_0^1 (x^2)(x^3) dx}{\int_0^1 (x^2)(x^2) dx} x^2 = x^3 - \frac{5}{6}x^2,$$

$$e_2 = \frac{f_2}{\sqrt{\int_0^1 f_2^2 dx}} = \frac{f_2}{\sqrt{\int_0^1 (x^6 + \frac{25}{36}x^4 - \frac{5}{3}x^5) dx}} = 6\sqrt{7} \left(x^3 - \frac{5}{6}x^2 \right).$$

Use equation 6.57 (i), we find that

$$\begin{aligned} p(x) &= \langle 2 + 3x, e_1 \rangle \cdot e_1 + \langle 2 + 3x, e_2 \rangle \cdot e_2 \\ &= \int_0^1 (2x^2 + 3x^3) dx \cdot 5x^2 + \int_0^1 \left(2x^3 + 3x^4 - \frac{5}{3}x^2 - \frac{5}{2}x^3 \right) dx \cdot 252 \left(x^3 - \frac{5}{6}x^2 \right) \\ &= \left(\frac{2}{3} + \frac{3}{4} \right) \cdot 5x^2 + 252 \left(\frac{2}{4} + \frac{3}{5} - \frac{5}{9} - \frac{5}{8} \right) \left(x^3 - \frac{5}{6}x^2 \right) \\ &= \frac{85}{12}x^2 - \frac{203}{10}x^3 + \frac{203}{12}x^2 \\ &= -20.3x^3 + 24x^2. \end{aligned}$$

Problem 3

Suppose $T \in \mathcal{L}(V)$ and U is a subspace of V .

- (a) Prove that if $U \subset \text{null } T$, then U is invariant under T .
- (b) Prove that if $\text{range } T \subset U$, then U is invariant under T .

✓ **Proof:** (a) If $U \subset \text{null } T$, then for every $u \in U$, we have $T(u) = 0$. $0 \in U$, so U is invariant under T .
 (b) For every $u \in U$, we have $T(u) = v$, where $v \in \text{range } T$. Because $\text{range } T \subset U$, then $v \in U$, so U is invariant under T . ■

Problem 4

Suppose $S, T \in \mathcal{L}(V)$ are such that $ST = TS$. Prove that $\text{range } S$ is invariant under T .

✓ **Proof:** For every $u \in \text{range } S$, we can find a $v \in V$ such that $S(v) = u$. Then

$$T(u) = T(S(v)) = TS(v) = ST(v) = S(T(v)),$$

which implies that $T(u) \in \text{range } S$. Thus $\text{range } S$ is invariant under T . ■

Problem 5

Suppose $S, T \in \mathcal{L}(V)$ are such that $ST = TS$. Prove that $\text{null } S$ is invariant under T .

✓ **Proof:** For every $u \in \text{null } S$, we have $S(u) = 0$. Then

$$ST(u) = TS(u) = T(S(u)) = T(0) = 0,$$

which implies that $T(u) \in \text{null } S$. Thus $\text{null } S$ is invariant under T . ■

Problem 6

Define $T \in \mathcal{L}(\mathbb{F}^3)$ by

$$T(z_1, z_2, z_3) = (2z_2, 0, 5z_3)$$

Find all eigenvalues and eigenvectors of T .

✓ **Solution:** With respect to the standard basis, the matrix of T is

$$A = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

Since A is upper triangular, its diagonal entries are the eigenvalues (by Theorem 5.41). Hence the eigenvalues of T are 0 (algebraic multiplicity 2) and 5 (algebraic multiplicity 1).

Eigenvalue $\lambda = 0$: We solve $A\mathbf{x} = \mathbf{0}$ for $\mathbf{x} = (x_1, x_2, x_3)^T \in \mathbb{F}^3$:

$$\begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \implies \begin{cases} 2x_2 = 0, \\ 0 = 0, \\ 5x_3 = 0. \end{cases}$$

Thus $x_2 = x_3 = 0$, while x_1 is free. Every solution is of the form $(x_1, 0, 0)^T$, so a basis for the eigenspace is $\{(1, 0, 0)^T\}$. Converting back to $\mathbb{F}^3$ under the standard basis, the eigenvectors of T for $\lambda = 0$ are all nonzero scalar multiples $k(1, 0, 0)$ with $k \in \mathbb{F} \setminus \{0\}$.

Eigenvalue $\lambda = 5$: We solve $(A - 5I)\mathbf{x} = \mathbf{0}$, where I is the 3×3 identity matrix. First compute

$$A - 5I = \begin{pmatrix} -5 & 2 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The system $(A - 5I)\mathbf{x} = \mathbf{0}$ gives

$$\begin{cases} -5x_1 + 2x_2 = 0, \\ -5x_2 = 0, \\ 0 = 0. \end{cases}$$

From the second equation $x_2 = 0$. Substituting into the first gives $x_1 = 0$. The third equation places no restriction on x_3 . Hence every solution is $(0, 0, x_3)^T$, and a basis for the eigenspace is $\{(0, 0, 1)^T\}$. Therefore the eigenvectors of T for $\lambda = 5$ are all nonzero scalar multiples $k(0, 0, 1)$, $k \in \mathbb{F} \setminus \{0\}$.

Problem 7

Define $T : \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$ by $Tp = p'$. Find all eigenvalues and eigenvectors of T .

✓ **Solution:** $(x^n)' = nx^{n-1}$. Suppose λ is an eigenvalue of T with an eigenvector q , then $q' = Tq = \lambda q$. If $\lambda \neq 0$, then $\deg \lambda q > \deg q'$, we get a contradiction. If $\lambda = 0$, the equation becomes $q' = 0$, and thus we can take $q = c$ for nonzero $c \in \mathbb{R}$. Hence the only eigenvalue of T is 0 with nonzero constant polynomials as eigenvectors.

! **Note 7.1:** We cannot use matrix on infinite-dimensional vector spaces.

Problem 8

Suppose $T \in \mathcal{L}(V)$. Suppose $S \in \mathcal{L}(V)$ is invertible.

(a) Prove that T and $S^{-1}TS$ have the same eigenvalues.

(b) What is the relationship between the eigenvectors of T and the eigenvectors of $S^{-1}TS$?

✓ **Proof:** (a) Since S is invertible, we have $I = S^{-1}S$, so

$$S^{-1}TS - \lambda I = S^{-1}(T - \lambda I)S.$$

If λ is an eigenvalue of T , there exists $x \neq 0$ such that $(T - \lambda I)x = 0$. Let $y = S^{-1}x \neq 0$. Then

$$(S^{-1}TS - \lambda I)y = S^{-1}(T - \lambda I)SS^{-1}x = S^{-1}(T - \lambda I)x = 0,$$

so λ is an eigenvalue of $S^{-1}TS$ with y as the corresponding eigenvector.

Conversely, if λ is an eigenvalue of $S^{-1}TS$, there exists $x \neq 0$ such that $(S^{-1}TS - \lambda I)x = 0$. Let $y = Sx \neq 0$. Then

$$(T - \lambda I)y = S(S^{-1}TS - \lambda I)x = 0,$$

so λ is an eigenvalue of T with y as the corresponding eigenvector.

Thus T and $S^{-1}TS$ share exactly the same eigenvalues.

(b) Let $\lambda \in \mathbb{F}$ be an eigenvalue of T . As we showed in part (a), this is the case if and only if λ is an eigenvalue of $S^{-1}TS$. Define

$$E(\lambda, T) = \{v \in V : v \neq 0, Tv = \lambda v\} \quad \text{and} \quad E(\lambda, S^{-1}TS) = \{u \in V : u \neq 0, (S^{-1}TS)(u) = \lambda u\}.$$

That is, $E(\lambda, T)$ is the collection of eigenvectors of T corresponding to the eigenvalue λ and $E(\lambda, S^{-1}TS)$ is the collection of eigenvectors of $S^{-1}TS$ corresponding to the eigenvalue λ . Our calculations in part (a) show that

$$E(\lambda, T) = \{Su : u \in E(\lambda, S^{-1}TS)\} \quad \text{and} \quad E(\lambda, S^{-1}TS) = \{S^{-1}v : v \in E(\lambda, T)\}. \quad \blacksquare$$

Problem 9

Find all eigenvalues and eigenvectors of the backward shift operator $T \in \mathcal{L}(\mathbb{F}^\infty)$ defined by

$$T(z_1, z_2, z_3, \dots) = (z_2, z_3, \dots).$$

✓ **Solution:** Suppose $\lambda \in \mathbb{F}$ is an eigenvalue of T and $z = (z_1, z_2, \dots) \neq 0$ is a corresponding eigenvector. The equation $Tz = \lambda z$ gives

$$(z_2, z_3, z_4, \dots) = (\lambda z_1, \lambda z_2, \lambda z_3, \dots),$$

which yields the recurrence

$$z_{k+1} = \lambda z_k \quad (k \geq 1).$$

Solving recursively, we obtain $z_{k+1} = \lambda^k z_1$ for all $k \geq 1$, where z_1 can be chosen arbitrarily. Because an eigenvector must be non-zero, we require $z_1 \neq 0$.

Thus every $\lambda \in \mathbb{F}$ is an eigenvalue of T , and the corresponding eigenvectors are

$$\{(w, \lambda w, \lambda^2 w, \dots) \mid w \in \mathbb{F} \setminus \{0\}\}.$$

Each eigenspace is one-dimensional, spanned by the vector $(1, \lambda, \lambda^2, \dots)$.

! **Note 9.1:** We cannot use matrix on infinite-dimensional vector spaces.

Problem 10

If A is a matrix with $m \times n$ dimension, please show that $A^T A$ and AA^T have the same nonzero eigenvalues.

✓ **Proof:** Suppose x is an eigenvector for a nonzero eigenvalue λ of $A^T A$. Then $A^T A x = \lambda x$. Multiplying both sides on the left by A gives

$$AA^T(Ax) = \lambda(Ax).$$

If $Ax = 0$, then $A^T A x = 0 = \lambda x$, contradicting $\lambda \neq 0$ and $x \neq 0$. Hence $Ax \neq 0$, and λ is also an eigenvalue of AA^T with eigenvector Ax .

Conversely, let x be an eigenvector for a nonzero eigenvalue λ of AA^T . Then $AA^T x = \lambda x$. Multiplying by A^T on the left yields

$$A^T A(A^T x) = \lambda(A^T x).$$

If $A^T x = 0$, then $AA^T x = 0 = \lambda x$, again a contradiction. Thus $A^T x \neq 0$, and λ is an eigenvalue of $A^T A$ with eigenvector $A^T x$. This completes the proof. ■

✓ **Proof based on SVD:** Assume A is a real matrix (the complex case is handled by using conjugate transposes). Let

$$A = U\Sigma V^T$$

be a singular value decomposition, where $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ are orthogonal matrices, and $\Sigma \in \mathbb{R}^{m \times n}$ is diagonal with nonnegative entries $\sigma_1, \dots, \sigma_r$ ($r = \text{rank } A$) on the diagonal, and zeros elsewhere. Then

$$A^T A = V\Sigma^T \Sigma V^T, \quad AA^T = U\Sigma \Sigma^T U^T.$$

Both $\Sigma^T \Sigma$ and $\Sigma \Sigma^T$ are diagonal matrices: the first has diagonal entries $\sigma_1^2, \dots, \sigma_r^2$ followed by $n - r$ zeros; the second has $\sigma_1^2, \dots, \sigma_r^2$ followed by $m - r$ zeros. Since V and U are orthogonal, $A^T A$ is orthogonally similar to $\Sigma^T \Sigma$, and AA^T is orthogonally similar to $\Sigma \Sigma^T$. Consequently, their eigenvalues are exactly the diagonal entries of those matrices. The nonzero eigenvalues of both products are therefore exactly $\sigma_1^2, \dots, \sigma_r^2$, which coincide. ■